

graph-theory-symbol— A set of LaTeX commands to have symbols designed for Graph Theory

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? ?

Abstract

This documentation is for package GraphTheorySymbol.

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1 Usage

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call. $\{\langle commandname \rangle\}$ The name of the command to create. $\{\langle commandcode \rangle\}$ The code of the command to create. This macro does nothing. It is merely an example. If this were a real macro, you would put a paragraph here describing what the macro is supposed to do, what its mandatory and optional arguments are, and so forth. Some text about an example macro called `\examplemacro`, which might have an optional argument $[\langle arg1 \rangle]$ and mandatory one $\{\langle arg2 \rangle\}$.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”. $\{\langle color \rangle\}$ The color that must be used by GraphTheorySymbol afterward.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as $\text{Adj}(\dots)$.

2 Implementation

```
1 \<package>
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{fdsymbol}
6 \RequirePackage{lua-unicode-math}
7 \RequirePackage{xcolor}
8
9 \def \GTS@mainColor {black}
10 \def \GTS@secondColor {gray}
11 \def \GTS@mainColorBackup {black}
12 \def \GTS@secondColorBackup {gray}
13 \def \GTS@drawSlash {}
14 \def \GTS@drawUpArrow {}
```

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```

15 \def \GTS@drawDownArrow {}
16
\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.  $\langle commandname \rangle$  The name of the command to
create.  $\langle commandcode \rangle$  The code of the command to create.
17 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
18   \@ifdefinable{#1}{\def#1/{#2}}
19 }}

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value
“black”.  $\langle color \rangle$  The color that must be used by GraphTheorySymbol afterward.
20 \newcommand{\GTSsetMainColor}[1]{%
21 \def\GTS@mainColorBackup{\GTS@mainColor}%
22 \def\GTS@mainColor{#1}%
23 }

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial
value “gray”.  $\langle color \rangle$  The color that must be used by GraphTheorySymbol afterward.
24 \newcommand{\GTSsetSecondColor}[1]{%
25 \def\GTS@secondColorBackup{\GTS@secondColor}%
26 \def\GTS@secondColor{#1}%
27 }

\GTSadjacency This macro will draw an adjacency symbol.
\GTSa 28 \GTS@newcommand{\GTSadjacency}{%
29 \text{\ensuremath{%
30 \raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
31 %
32 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
33 $\mathcolor{\GTS@secondColor}{\circ}$%
34 }}%
35 \hspace{-0.39ex}%
36 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
37 $\mathcolor{\GTS@secondColor}{\bullet}$%
38 }}%
39 \hspace{-0.12ex}%
40 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
41 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
42 }}%
43 \hspace{-0.40ex}%
44 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
45 $\mathcolor{\GTS@secondColor}{\circ}$%
46 }}%
47 \hspace{-0.39ex}%
48 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
49 $\mathcolor{\GTS@secondColor}{\bullet}$%
50 }}%
51 \hspace{0.27ex}%
52 \if\GTS@drawUpArrow\empty\else%
53 \hspace{-0.65ex}%
54 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
55 \textcolor{\GTS@mainColor}{\textasciicircum}%
56 }}%
57 \fi%
58 \if\GTS@drawDownArrow\empty\else%
59 \hspace{-0.65ex}%
60 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
61 \textcolor{\GTS@mainColor}{\textasciicircum}%
62 }}%
63 \fi%
64 \if\GTS@drawSlash\empty\else%
65 \hspace{-0.65ex}%
66 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
67 \textcolor{black}{/}%
68 }}%

```

```

69 \fi%
70 %
71 }}%
72 }}%
73 }
74
75 \GTS@newcommand{\GTSa}{\GTSadjacency/}

\GTSadjacencyRelation This macro will draw an adjacency symbol with relation spacing in math mode.
\GTSaR 76 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSadjacency/}}
77 \GTS@newcommand{\GTSaR}{\GTSadjacencyRelation/}

\GTSunadjacency This macro will draw an unadjacency symbol.
\GTSu 78 \GTS@newcommand{\GTSunadjacency}{%
79 \def\GTS@drawSlash{1}%
80 \GTSadjacency/%
81 \def\GTS@drawSlash{}%
82 }
83
84 \GTS@newcommand{\GTSu}{\GTSunadjacency/}

\GTSunadjacencyRelation This macro will draw an unadjacency symbol with relation spacing in math mode.
\GTSuR 85 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSunadjacency/}}
86 \GTS@newcommand{\GTSuR}{\GTSunadjacencyRelation/}

\GTSdirectedAdjacencyUp This macro will draw a directed adjacency upward symbol. The operand on the left of the
\GTSdAU symbol should be considered the source of the arc; the operand on the right of the symbol
should be considered the target of the arc.
87 \GTS@newcommand{\GTSdirectedAdjacencyUp}{%
88 \def\GTS@drawUpArrow{1}%
89 \GTSadjacency/%
90 \def\GTS@drawUpArrow{}%
91 }
92
93 \GTS@newcommand{\GTSdAU}{\GTSdirectedAdjacencyUp/}

\GTSdirectedAdjacencyUpRelation This macro will draw a directed adjacency upward symbol with relation spacing in math
\GTSdAUR mode.
94 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{%
95 \mathrel{\GTSdirectedAdjacencyUp/}%
96 }
97 \GTS@newcommand{\GTSdAUR}{\GTSdirectedAdjacencyUpRelation/}

\GTSnegatedDirectedAdjacencyUp This macro will draw a negated directed adjacency upward symbol.
\GTSnDAU 98 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{%
99 \def\GTS@drawSlash{1}%
100 \GTSdirectedAdjacencyUp/%
101 \def\GTS@drawSlash{}%
102 }
103
104 \GTS@newcommand{\GTSnDAU}{\GTSnegatedDirectedAdjacencyUp/}

\GTSnegatedDirectedAdjacencyUpRelation This macro will draw a negated directed adjacency upward symbol with relation spacing in
\GTSnDAUR math mode.
105 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
106 \mathrel{\GTSnegatedDirectedAdjacencyUp/}%
107 }
108 \GTS@newcommand{\GTSnDAUR}{\GTSnegatedDirectedAdjacencyDownRelation/}

\GTSdirectedAdjacencyDown This macro will draw a directed adjacency downward symbol. The operand on the left of the
\GTSdAD symbol should be considered the target of the arc; the operand on the right of the symbol
should be considered the source of the arc.
109 \GTS@newcommand{\GTSdirectedAdjacencyDown}{%
110 \def\GTS@drawDownArrow{1}%
111 \GTSadjacency/%

```

```

112 \def\GTS@drawDownArrow{%
113 }
114
115 \GTS@newcommand{\GTSdAD}{\GTSdirectedAdjacencyDown/}
directedAdjacencyDownRelation This macro will draw a directed adjacency downward symbol with relation spacing in math
\GTSdADR mode.
116 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{%
117 \mathrel{\GTSdirectedAdjacencyDown/}%
118 }
119 \GTS@newcommand{\GTSdADR}{\GTSdirectedAdjacencyDownRelation/}
\GTSnegatedDirectedAdjacencyDown This macro will draw a negated directed adjacency downward symbol.
\GTSnDAD 120 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{%
121 \def\GTS@drawSlash{1}%
122 \GTSdirectedAdjacencyDown/%
123 \def\GTS@drawSlash{}%
124 }
125
126 \GTS@newcommand{\GTSnDAD}{\GTSnegatedDirectedAdjacencyDown/}
DirectedAdjacencyDownRelation This macro will draw a negated directed adjacency downward symbol with relation spacing
\GTSnDADR in math mode.
127 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
128 \mathrel{\GTSnegatedDirectedAdjacencyDown/}%
129 }
130 \GTS@newcommand{\GTSnDADR}{\GTSnegatedDirectedAdjacencyDownRelation/}

\GTSincidence This macro will draw an incidence symbol. The operand on the left of the symbol should be
\GTSi considered the vertex to which the edge/the operand on the right of the symbol is incident.
131 \GTS@newcommand{\GTSincidence}{%
132 \text{\ensuremath{%
133 \raisebox{0ex}{\scalebox{1.8}[1.8]{%
134 %
135 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
136 $\mathcolor{\GTS@secondColor}{\circ}$%
137 } }%
138 \hspace{-0.39ex}%
139 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
140 $\mathcolor{\GTS@secondColor}{\bullet}$%
141 } }%
142 \hspace{-0.12ex}%
143 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
144 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
145 } }%
146 \hspace{-0.18ex}%
147 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
148 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
149 } }%
150 \hspace{0.2ex}%
151 \if\GTS@drawUpArrow\empty\else%
152 \hspace{-0.65ex}%
153 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
154 \textcolor{\GTS@secondColor}{\textasciicircum}%
155 } }%
156 \fi%
157 \if\GTS@drawDownArrow\empty\else%
158 \hspace{-0.65ex}%
159 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
160 \textcolor{\GTS@secondColor}{\textasciicircum}%
161 } }%
162 \fi%
163 \if\GTS@drawSlash\empty\else%
164 \hspace{-0.65ex}%
165 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%

```

```

166 \textcolor{black}{/}%
167 }}%
168 \fi%
169 %
170 }}%
171 }}%
172 }
173
174 \GTS@newcommand{\GTSi}{\GTSincidence/}

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.
\GTSiR 175 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSincidence/}}
176 \GTS@newcommand{\GTSiR}{\GTSincidenceRelation/}

\GTSnegatedIncidence This macro will draw a negated incidence symbol.
\GTSnI 177 \GTS@newcommand{\GTSnegatedIncidence}{%
178 \def\GTS@drawSlash{1}%
179 \GTSincidence/%
180 \def\GTS@drawSlash{}%
181 }
182
183 \GTS@newcommand{\GTSnI}{\GTSnegatedIncidence/}

\GTSnegatedIncidenceRelation This macro will draw a negated incidence symbol with relation spacing in math mode.
\GTSnIR 184 \GTS@newcommand{\GTSnegatedIncidenceRelation}{%
185 \mathrel{\GTSnegatedIncidence/}%
186 }
187 \GTS@newcommand{\GTSnIR}{\GTSnegatedIncidenceRelation/}

\GTSoutIncidence This macro will draw a directed incidence upward (out(ward) incidence) symbol. The
\GTSoI operand on the left of the symbol should be considered the vertex to which the arc/the
operand on the right of the symbol is incident.
188 \GTS@newcommand{\GTSoutIncidence}{%
189 \def\GTS@drawUpArrow{1}%
190 \GTSincidence/%
191 \def\GTS@drawUpArrow{}%
192 }
193
194 \GTS@newcommand{\GTSoI}{\GTSoutIncidence/}

\GTSoutIncidenceRelation This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation
\GTSoIR spacing in math mode.
195 \GTS@newcommand{\GTSoutIncidenceRelation}{%
196 \mathrel{\GTSoutIncidence/}%
197 }
198 \GTS@newcommand{\GTSoIR}{\GTSoutIncidenceRelation/}

\GTSnegatedOutIncidence This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.
\GTSnOI 199 \GTS@newcommand{\GTSnegatedOutIncidence}{%
200 \def\GTS@drawSlash{1}%
201 \GTSoutIncidence/%
202 \def\GTS@drawSlash{}%
203 }
204
205 \GTS@newcommand{\GTSnOI}{\GTSnegatedOutIncidence/}

\GTSnegatedOutIncidenceRelation This macro will draw a negated directed incidence upward (out(ward) incidence) symbol
\GTSnOIR with relation spacing in math mode.
206 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{%
207 \mathrel{\GTSnegatedOutIncidence/}%
208 }
209 \GTS@newcommand{\GTSnOIR}{\GTSnegatedOutIncidenceRelation/}

```

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

210 \GTS@newcommand{\GTSinIncidence}{%
211 \def\GTS@drawDownArrow{1}%
212 \GTSinIncidence/%
213 \def\GTS@drawDownArrow{}}%
214 }
215
216 \GTS@newcommand{\GTSiI}{\GTSinIncidence/}

```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

217 \GTS@newcommand{\GTSinIncidenceRelation}{%
218 \mathrel{\GTSinIncidence/}%
219 }
220 \GTS@newcommand{\GTSiIR}{\GTSinIncidenceRelation/}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```

221 \GTS@newcommand{\GTSnegatedInIncidence}{%
222 \def\GTS@drawSlash{1}%
223 \GTSinIncidence/%
224 \def\GTS@drawSlash{}}%
225 }
226
227 \GTS@newcommand{\GTSnII}{\GTSnegatedInIncidence/}

```

`\GTSnegatedInIncidenceRelation` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

228 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{%
229 \mathrel{\GTSnegatedInIncidence/}%
230 }
231 \GTS@newcommand{\GTSnIIR}{\GTSnegatedInIncidenceRelation/}

```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

```

232 \GTS@newcommand{\GTSadjacencyMonochrome}{%
233 \GTSsetMainColor{black}%
234 \GTSadjacency/%
235 \GTSsetMainColor{\GTS@mainColorBackup}%
236 }
237
238 \GTS@newcommand{\GTSaM}{\GTSadjacencyMonochrome/}

```

`\GTSadjacencyMonochromeRelation` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

```

239 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{%
240 \mathrel{\GTSadjacencyMonochrome/}%
241 }
242 \GTS@newcommand{\GTSaMR}{\GTSadjacencyMonochromeRelation/}

```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

```

243 \GTS@newcommand{\GTSunadjacencyMonochrome}{%
244 \def\GTS@drawSlash{1}%
245 \GTSadjacencyMonochrome/%
246 \def\GTS@drawSlash{}}%
247 }
248
249 \GTS@newcommand{\GTSuM}{\GTSunadjacencyMonochrome/}

```

`\GTSunadjacencyMonochromeRelation` This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

```

250 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{%
251 \mathrel{\GTSunadjacencyMonochrome/}%
252 }
253 \GTS@newcommand{\GTSuMR}{\GTSunadjacencyMonochromeRelation/}

```

directedAdjacencyUpMonochrome This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

\GTSdAUM
254 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
255 \def\GTS@drawUpArrow{1}%
256 \GTSadjacencyMonochrome/%
257 \def\GTS@drawUpArrow{}}%
258 }
259
260 \GTS@newcommand{\GTSdAUM}{\GTSdirectedAdjacencyUpMonochrome/}

```

AdjacencyUpMonochromeRelation This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

```

\GTSdAUMR
261 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
262 \mathrel{\GTSdirectedAdjacencyUpMonochrome/}%
263 }
264 \GTS@newcommand{\GTSdAUMR}{\GTSdirectedAdjacencyUpMonochromeRelation/}

```

DirectedAdjacencyUpMonochrome This macro will draw a monochrome negated directed adjacency upward symbol.

```

\GTSnDAUM
265 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
266 \def\GTS@drawSlash{1}%
267 \GTSdirectedAdjacencyUpMonochrome/%
268 \def\GTS@drawSlash{}}%
269 }
270
271 \GTS@newcommand{\GTSnDAUM}{\GTSnegatedDirectedAdjacencyUpMonochrome/}

```

AdjacencyUpMonochromeRelation This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```

\GTSnDAUMR
272 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
273 \mathrel{\GTSnegatedDirectedAdjacencyUpMonochrome/}%
274 }
275 \GTS@newcommand{\GTSnDAUMR}{%
276 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/}
277 }

```

directedAdjacencyDownMonochrome This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

\GTSdADM
278 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
279 \def\GTS@drawDownArrow{1}%
280 \GTSadjacencyMonochrome/%
281 \def\GTS@drawDownArrow{}}%
282 }
283
284 \GTS@newcommand{\GTSdADM}{\GTSdirectedAdjacencyDownMonochrome/}

```

AdjacencyDownMonochromeRelation This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdADMR
285 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
286 \mathrel{\GTSdirectedAdjacencyDownMonochrome/}%
287 }
288 \GTS@newcommand{\GTSdADMR}{\GTSdirectedAdjacencyDownMonochromeRelation/}

```

directedAdjacencyDownMonochrome This macro will draw a monochrome negated directed adjacency downward symbol.

```

\GTSnDADM
289 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
290 \def\GTS@drawSlash{1}%
291 \GTSdirectedAdjacencyDownMonochrome/%
292 \def\GTS@drawSlash{}}%
293 }
294
295 \GTS@newcommand{\GTSnDADM}{\GTSnegatedDirectedAdjacencyDownMonochrome/}

```

`\adjacencyDownMonochromeRelation` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```

296 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
297 \mathrel{\GTSnegatedDirectedAdjacencyDownMonochrome/}%
298 }
299 \GTS@newcommand{\GTSnDADMR}{%
300 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
301 }

```

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the `\GTSiM` symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

302 \GTS@newcommand{\GTSincidenceMonochrome}{%
303 \GTSsetMainColor{black}%
304 \GTSincidence/%
305 \GTSsetMainColor{\GTS@mainColorBackup}%
306 }
307
308 \GTS@newcommand{\GTSiM}{\GTSincidenceMonochrome/}

```

`\GTSincidenceMonochromeRelation` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```

309 \GTS@newcommand{\GTSincidenceMonochromeRelation}{%
310 \mathrel{\GTSincidenceMonochrome/}%
311 }
312 \GTS@newcommand{\GTSiMR}{\GTSincidenceMonochromeRelation/}

```

`\GTSnegatedIncidenceMonochrome` This macro will draw a monochrome negated incidence symbol.

```

313 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
314 \def\GTS@drawSlash{1}%
315 \GTSincidenceMonochrome/%
316 \def\GTS@drawSlash{}%
317 }
318
319 \GTS@newcommand{\GTSnIM}{\GTSnegatedIncidenceMonochrome/}

```

`\GTSnegatedIncidenceMonochromeRelation` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```

320 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{%
321 \mathrel{\GTSnegatedIncidenceMonochrome/}%
322 }
323 \GTS@newcommand{\GTSnIMR}{\GTSnegatedIncidenceMonochromeRelation/}

```

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

324 \GTS@newcommand{\GTSoutIncidenceMonochrome}{%
325 \def\GTS@drawUpArrow{1}%
326 \GTSincidenceMonochrome/%
327 \def\GTS@drawUpArrow{}%
328 }
329
330 \GTS@newcommand{\GTSoIM}{\GTSoutIncidenceMonochrome/}

```

`\GTSoutIncidenceMonochromeRelation` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

331 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{%
332 \mathrel{\GTSoutIncidenceMonochrome/}%
333 }
334 \GTS@newcommand{\GTSoIMR}{\GTSoutIncidenceMonochromeRelation/}

```

`\GTSnegatedOutIncidenceMonochrome` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```

335 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
336 \def\GTS@drawSlash{1}%

```



```

337 \GTSoutIncidenceMonochrome/%
338 \def\GTS@drawSlash{}%
339 }
340
341 \GTS@newcommand{\GTSnOIM}{\GTSnegatedOutIncidenceMonochrome/}

\GTSoutIncidenceMonochromeRelation This macro will draw a monochrome negated directed incidence upward (out(ward) inci-
\GTSnOIM dence) symbol with relation spacing in math mode.

342 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
343 \mathrel{\GTSnegatedOutIncidenceMonochrome/}%
344 }
345 \GTS@newcommand{\GTSnOIMR}{\GTSnegatedOutIncidenceMonochromeRelation/}

\GTSinIncidenceMonochrome This macro will draw a monochrome directed incidence downward (in(ward) incidence) sym-
\GTSiIM bol. The operand on the left of the symbol should be considered the vertex to which the
arc/the operand on the right of the symbol is incident.

346 \GTS@newcommand{\GTSinIncidenceMonochrome}{%
347 \def\GTS@drawDownArrow{1}%
348 \GTSinIncidenceMonochrome/%
349 \def\GTS@drawDownArrow{}%
350 }
351
352 \GTS@newcommand{\GTSiIM}{\GTSinIncidenceMonochrome/}

\GTSinIncidenceMonochromeRelation This macro will draw a monochrome directed incidence downward (in(ward) incidence) sym-
\GTSiIMR bol with relation spacing in math mode.

353 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{%
354 \mathrel{\GTSinIncidenceMonochrome/}%
355 }
356 \GTS@newcommand{\GTSiIMR}{\GTSinIncidenceMonochromeRelation/}

\GTSnegatedInIncidenceMonochrome This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
\GTSnIIM dence) symbol.

357 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
358 \def\GTS@drawSlash{1}%
359 \GTSinIncidenceMonochrome/%
360 \def\GTS@drawSlash{}%
361 }
362
363 \GTS@newcommand{\GTSnIIM}{\GTSnegatedInIncidenceMonochrome/}

\GTSnegatedInIncidenceMonochromeRelation This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
\GTSnIIMR dence) symbol with relation spacing in math mode.

364 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
365 \mathrel{\GTSnegatedInIncidenceMonochrome/}%
366 }
367 \GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochromeRelation/}

\dummyMacro This is a dummy macro. If it did anything, we'd describe its implementation here. "
368 \newcommand{\dummyMacro}{

Don't use % to introduce a code comment within a macrocode environment. Instead, you
should typeset all of your comments with LaTeX—doing so gives much prettier results. For
comments within a macro/environment body, just do an \end{macrocode}, include some
commentary, and do another \begin{macrocode}. It's that simple.

369 }

370 \endinput
371 \</package>

```